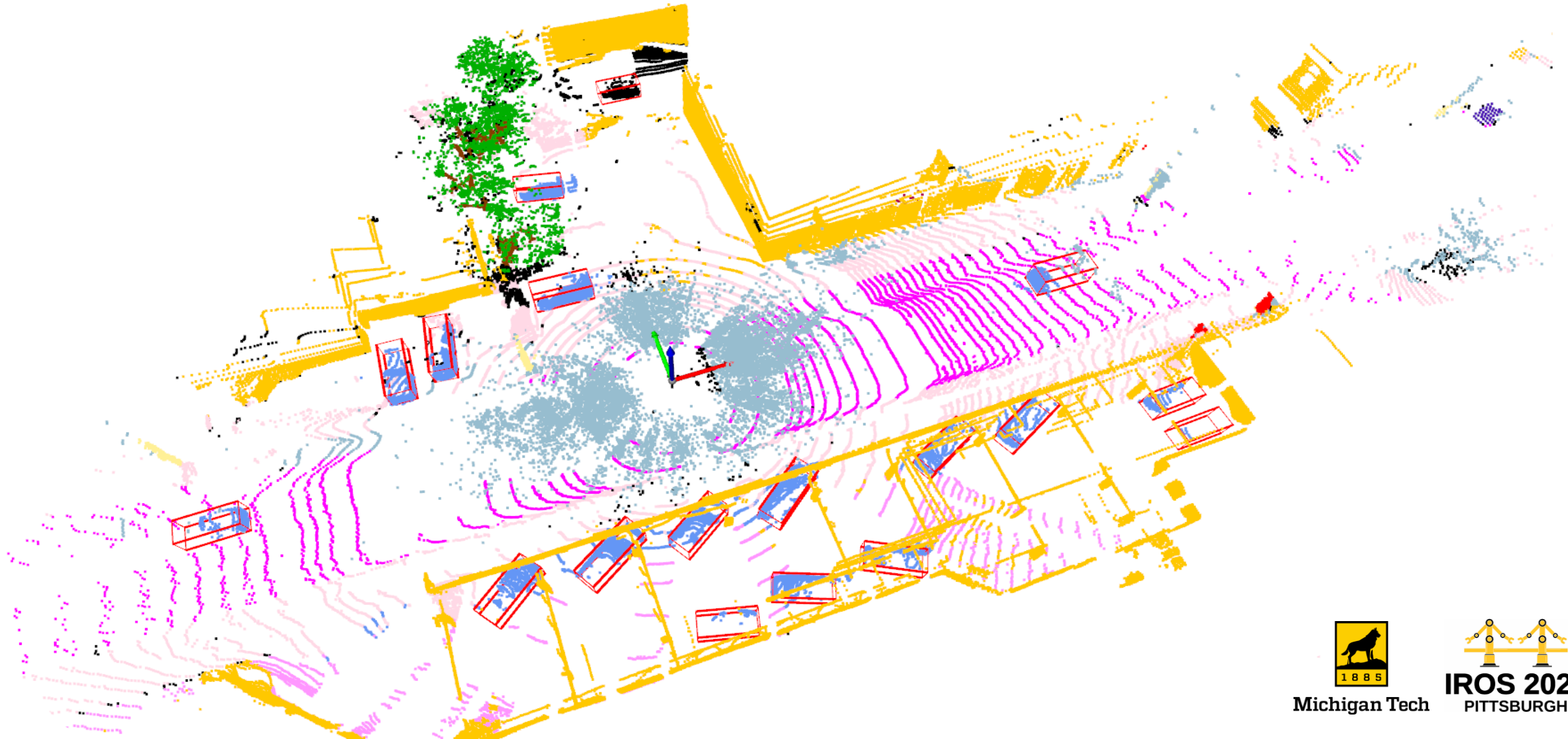
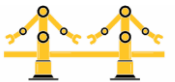


Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection

Yiming Yang & Jeremy P. Bos

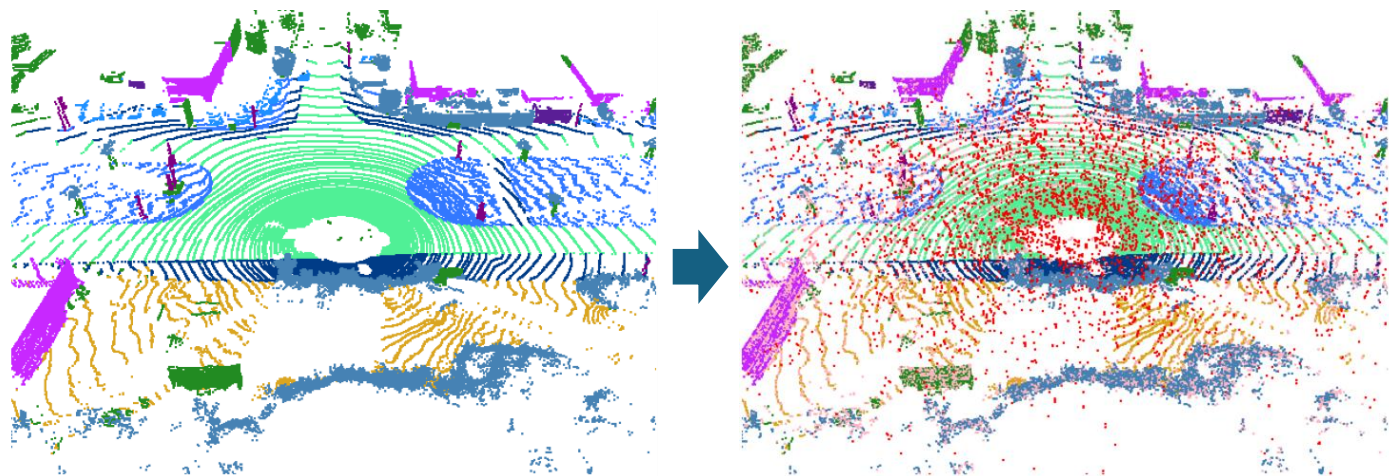


Michigan Tech



IROS 2026
PITTSBURGH

Existing research



Snow simulator

- Models airborne snow
- **No accumulated snow**

[Zhao et al. 2025]

Algorithm	Distance interval (m)	mAP-Car (%)		
		Sunny	Heavy snow	Difference
PV-RCNN	5 ~ 20	87.5	70.3	↓17.2
	20 ~ 30	77.4	62.8	↓14.6
	30 ~ 40	65.6	51.9	↓13.7
	40 ~ 50	57.2	43.3	↓13.9
Voxel-RCNN	5 ~ 20	89.2	71.6	↓17.6
	20 ~ 30	80.5	66.7	↓13.8
	30 ~ 40	69.7	56.3	↓13.4
	40 ~ 50	61.1	47.1	↓14.0
PointPillars	5 ~ 20	82.1	63.3	↓18.8
	20 ~ 30	72.9	58.0	↓14.9
	30 ~ 40	62.0	48.5	↓13.5
	40 ~ 50	53.4	39.3	↓14.1

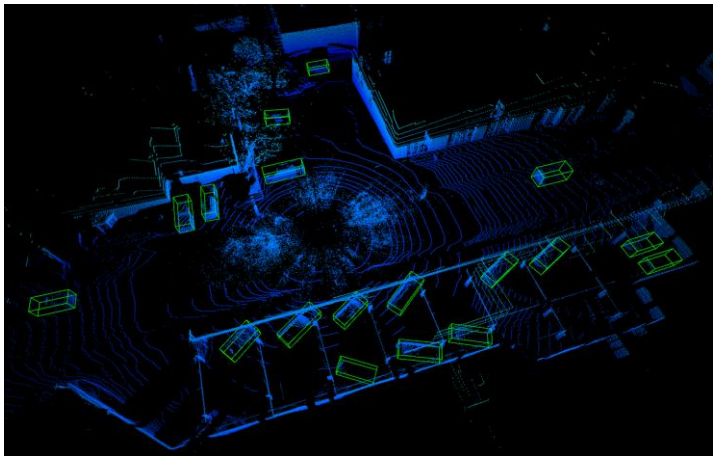
Real-world snow

- **Cross-Domain Variation**
- **Coarse category (light vs. heavy)**

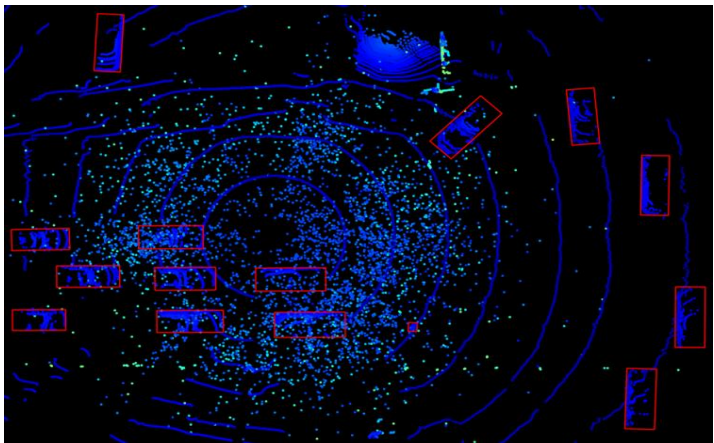
[Sun et al. 2024]

Proposed Approach

1. Two real-world large snow datasets, separate evaluation



WADS-3D:
2,050 labeled frames



CADC [Pitropov et al., 2020]:
7,000 labeled frames

2. 5 different neural detectors

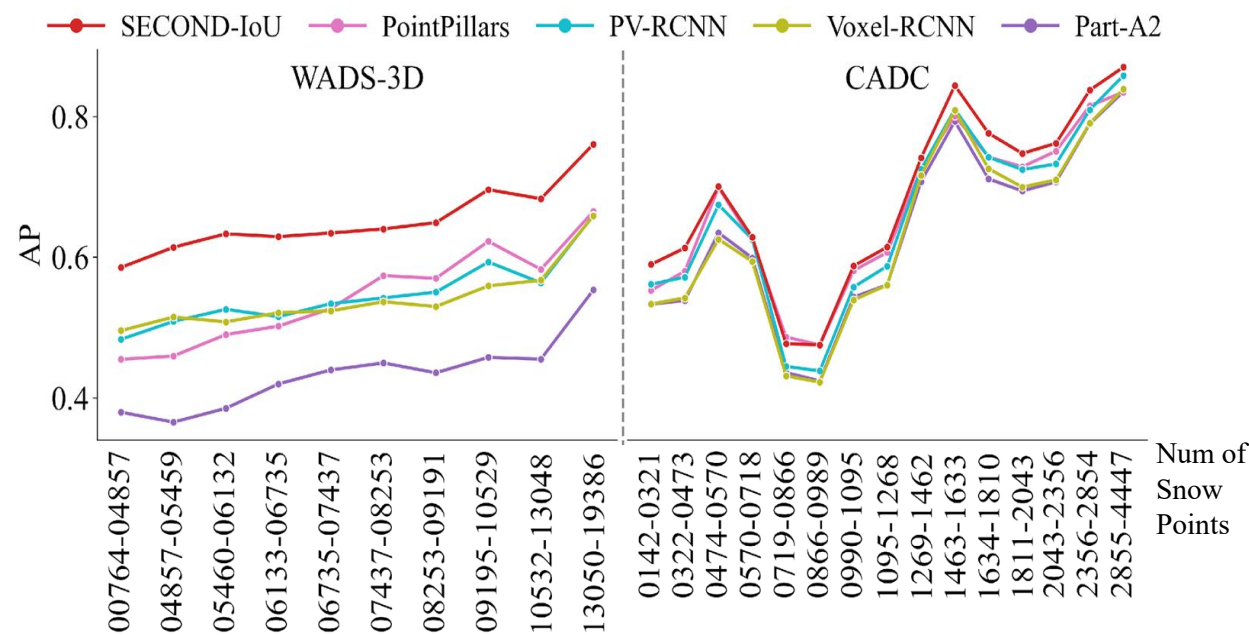
Detector	Architecture	Stages
SECOND-IOU	Voxel-based	One
Voxel R-CNN	Voxel-based	Two
PointPillar	Pillar-base	One
SECOND	Point-based	Two
PV-RCNN	Hybrid	Two

[OpenPCDet, 2020]

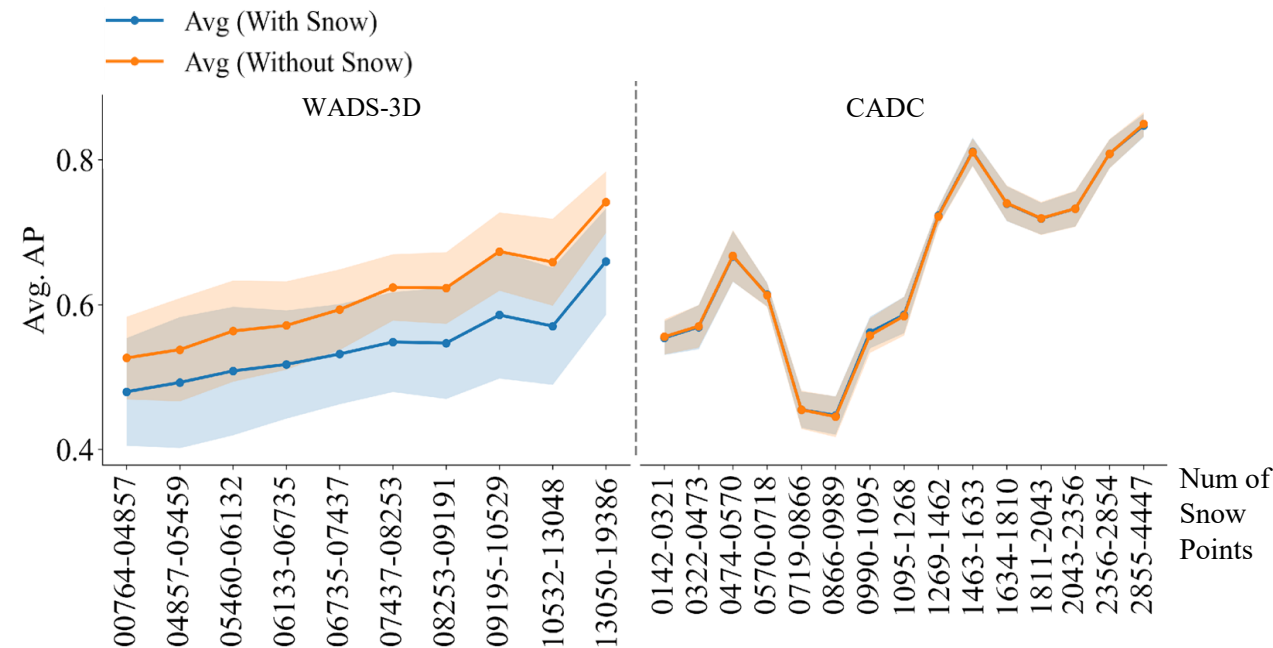
3. Granular snow metric:
number of raw snow points

Overall Performance

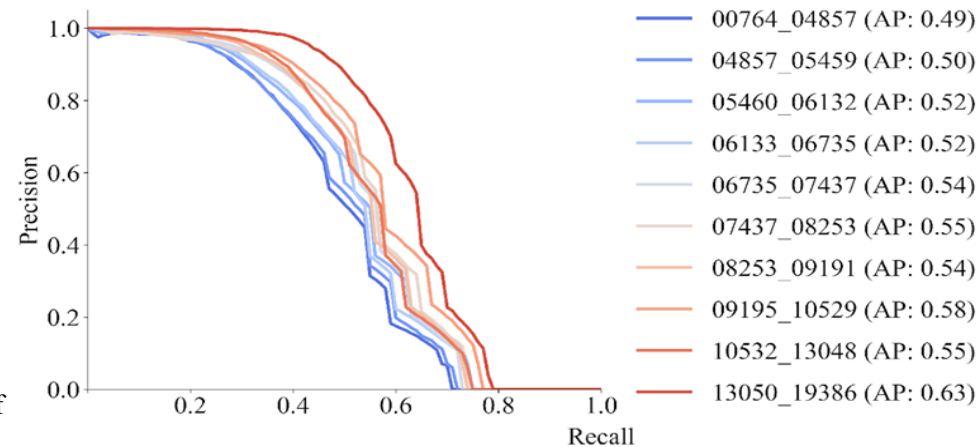
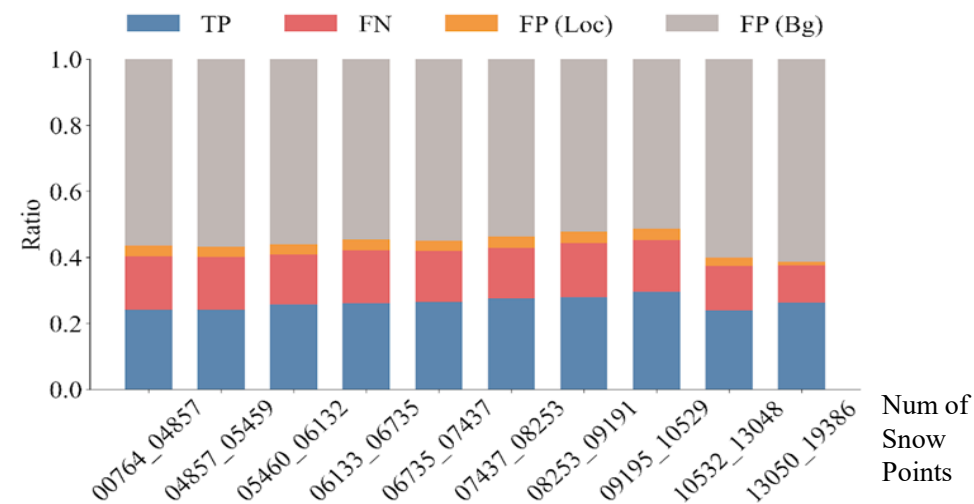
AP vs. Number of snow points



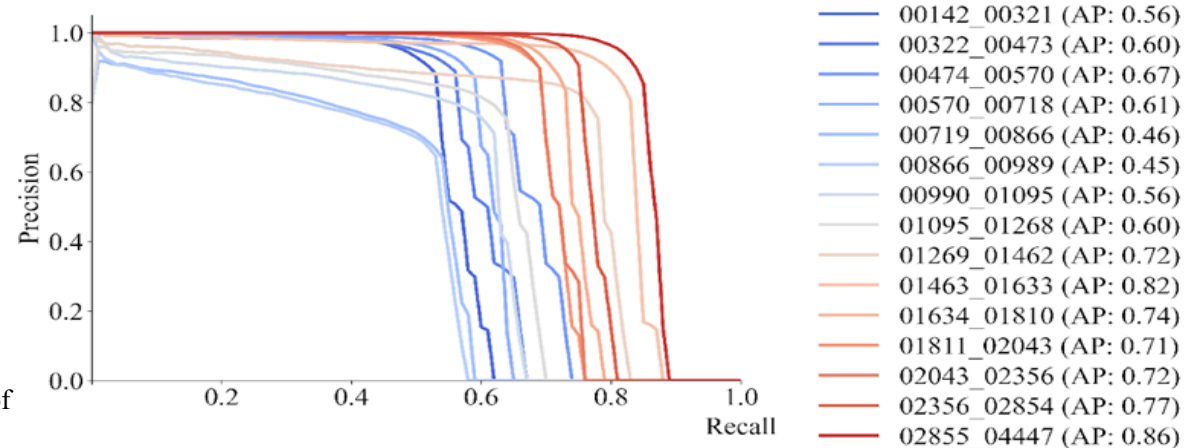
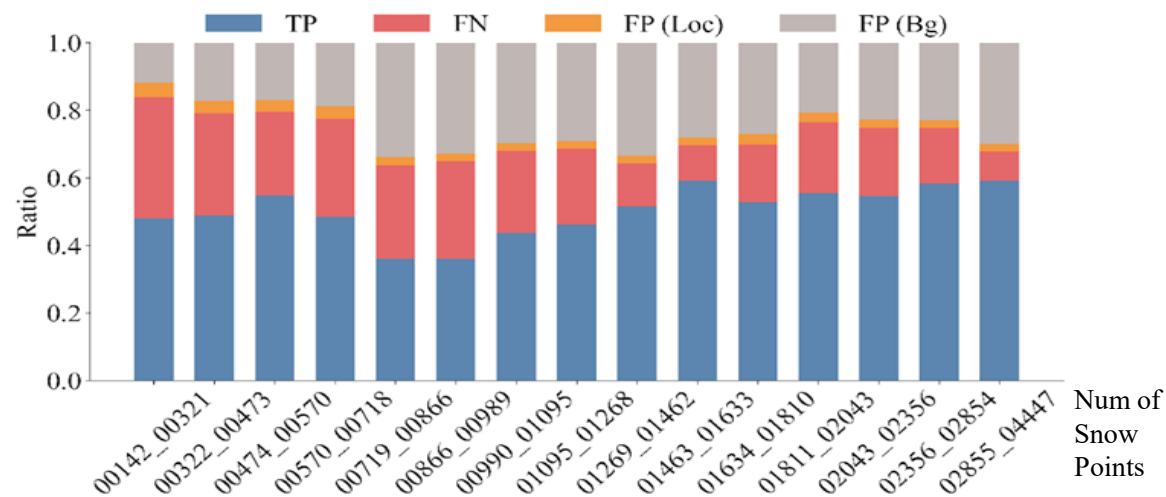
Ablation Study: with vs. without snow



Error Matrix and Recall Analysis

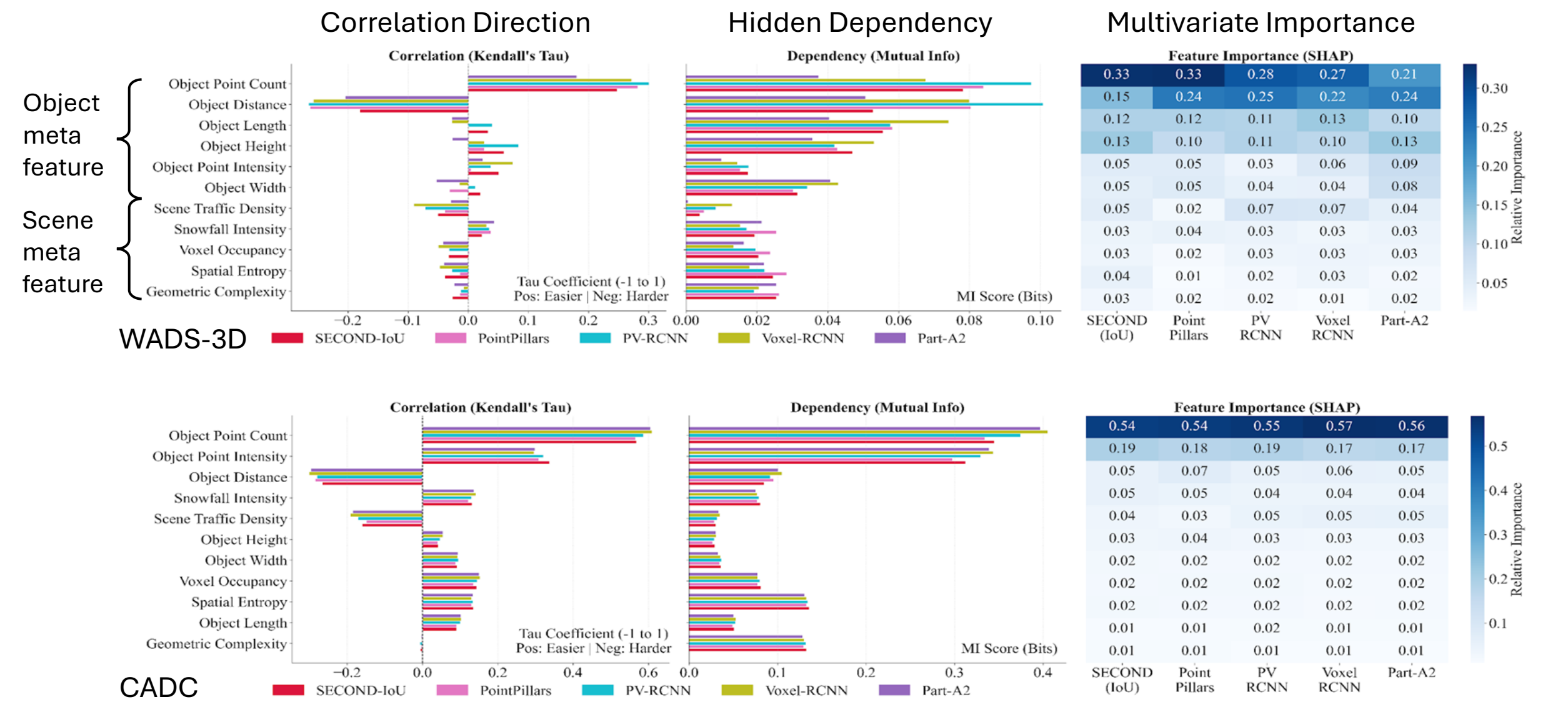


WADS-3D



CADC

Meta-analysis Results



Number of Snow Points vs. Key Meta Features

Dataset	Feature	Pearson r	p -value	Sig.
WADS-3D	Object Point Count	0.820	3.70×10^{-3}	**
	Object Distance	-0.282	0.430	NS
CADC	Object Point Count	0.733	1.86×10^{-3}	**
	Object Mean Intensity	0.694	4.12×10^{-3}	**

Bold values indicate statistical significance ($p < 0.05$).

Sig. levels: ** ($p < 0.01$), * ($p < 0.05$), NS (not significant).

Follow-up Study - Controlled Experiment

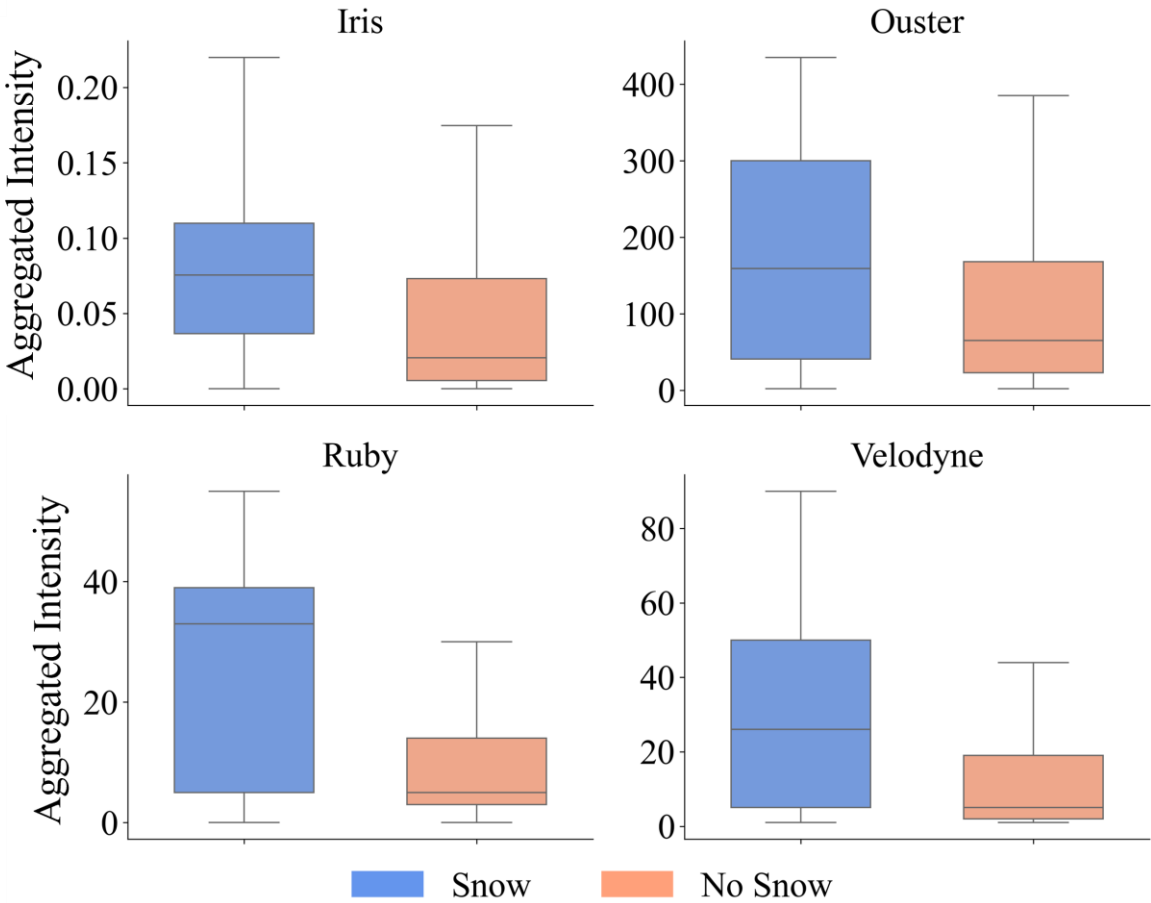
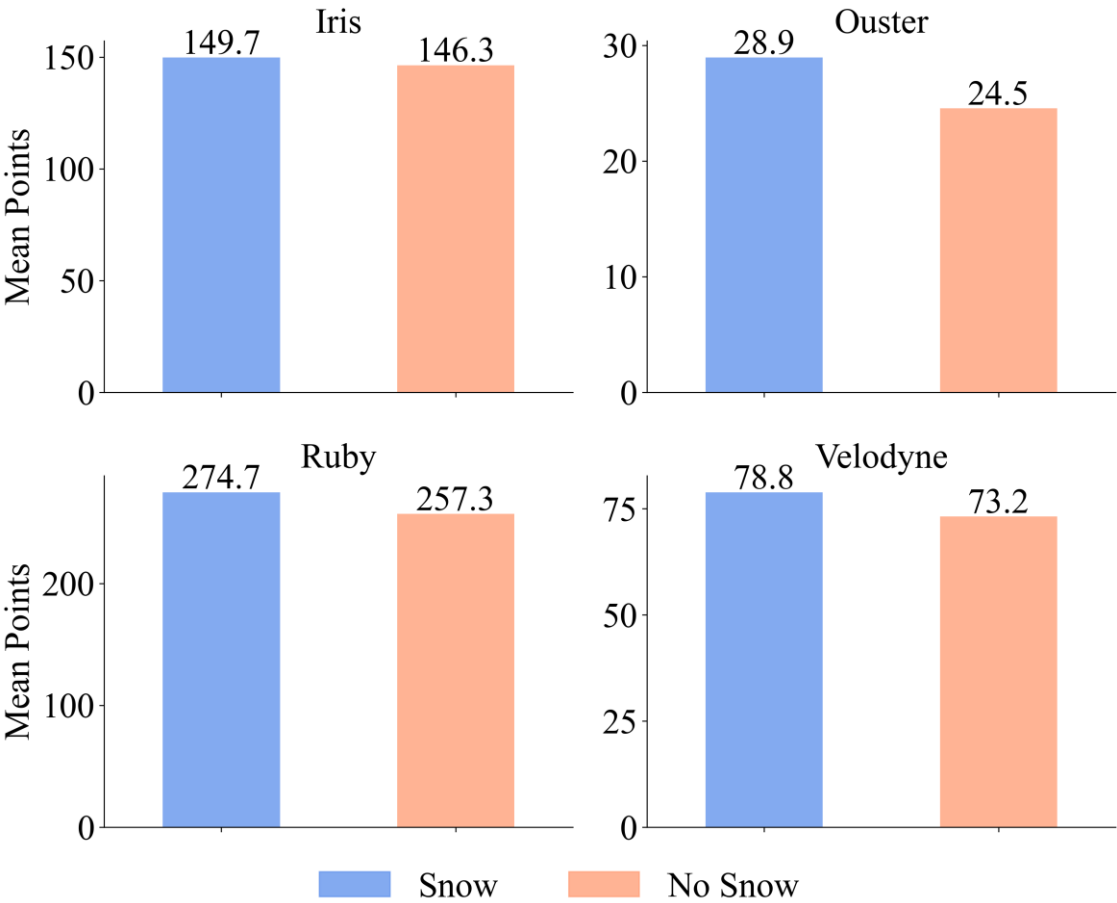


Manual Clean



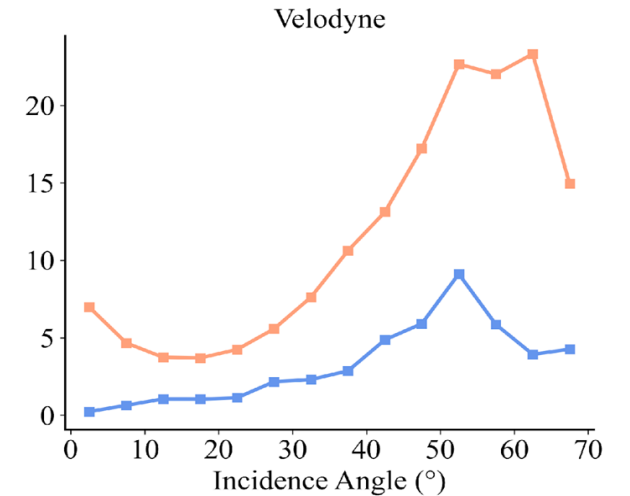
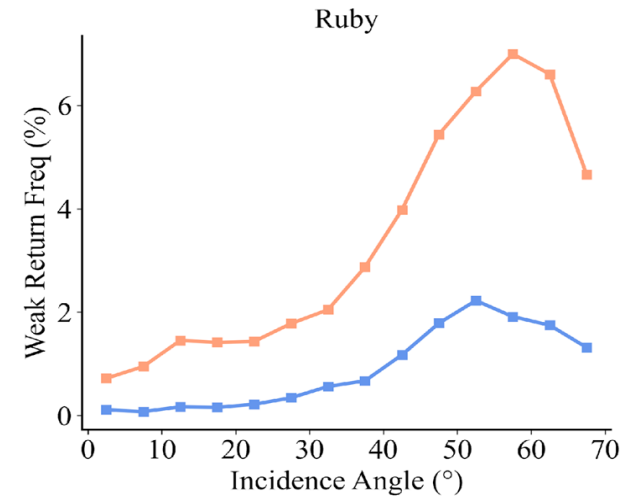
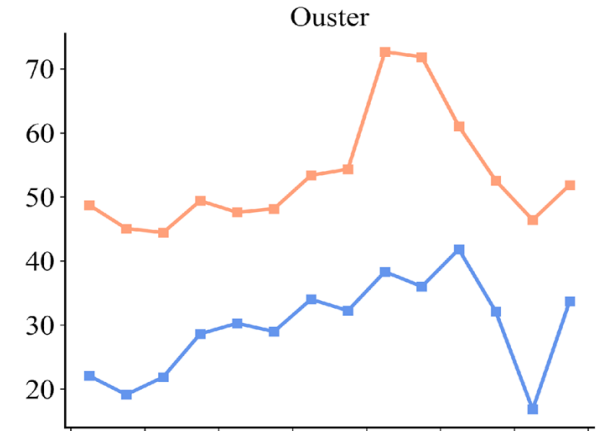
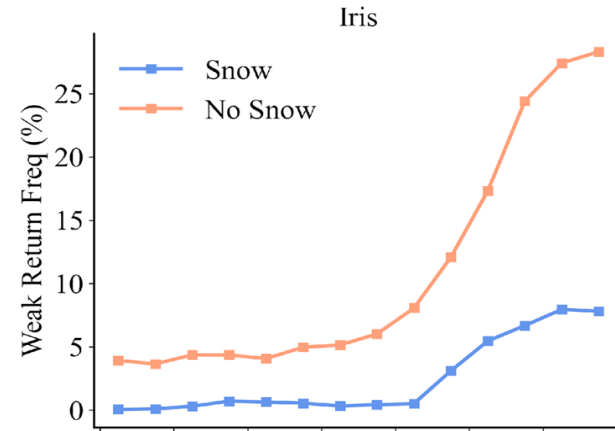
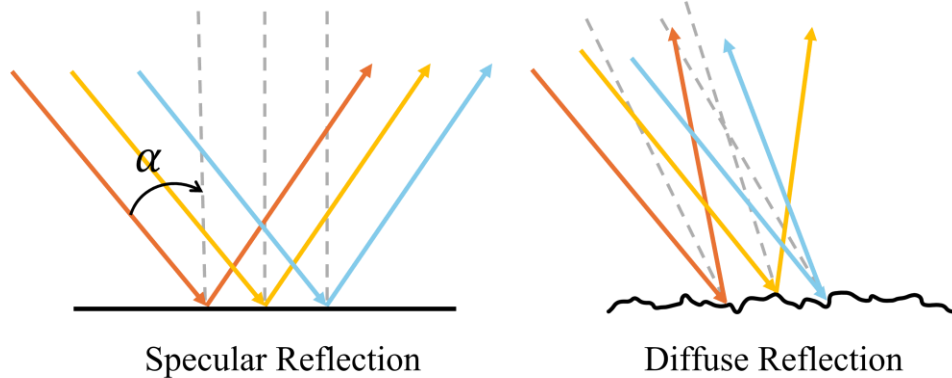
Parameter	Luminar Iris <i>(1550nm Raster)</i>	Robosense Ruby <i>(128-Channel)</i>	Ouster OS1-64 <i>(850nm Gen1)</i>	Velodyne VLP-32C <i>(32-Channel)</i>
Scanning Tech.	Raster Scan (MEMS)	Mechanical Spinner	Mechanical Spinner	Mechanical Spinner
Wavelength	1550 nm	905 nm	855 nm	903 nm
Channels	1 (Equivalent to 640 lines)	128	64	32
Vertical FOV	26° (0° to 26°)	40° (-25° to +15°)	33.2° (±16.6°)	40° (-25° to +15°)
Vertical Resolution	0.05° (Uniform)	0.1° (Non-uniform)	0.52° (Uniform)	0.33° (Non-uniform)
Horizontal FOV	120°	360°	360°	360°
Horizontal Resolution	0.05°	0.2°/0.4°	0.17° (at 1024 mode)	0.1° – 0.4°
Rotation / Scan Rate	1 – 30 fps	10 Hz / 20 Hz	10 Hz / 20 Hz	5 – 20 Hz
Raw Intensity	7-bit (0–127)	8-bit (0–255)	16-bit (0–65535)	8-bit (0–255)

Point Count & Intensity Distribution



Weak Return vs Angle of Incidence

- Weak returns: the first quartile (Q1) of intensity data
- Angle of incidence: the angle between the surface normal and the incoming laser beam



Conclusion

Accumulated snow change object characteristics:

- Increasing point count
- Raising point intensity
- Mitigating weak returns

Future Work

- Find detection threshold: Identify when dimension boost by snow harm detection performance.
- Test smaller targets: Evaluate detection on pedestrians and cyclists.

Thank you! Questions?

Contact Information

Yiming Yang (Presenter):
yiming.yang@okstate.edu

Jeremy P. Bos:
jpbos@mtu.edu

WADS Dataset:

<https://digitalcommons.mtu.edu/wads/>



LiDAR | RGB | Thermal | GNSS

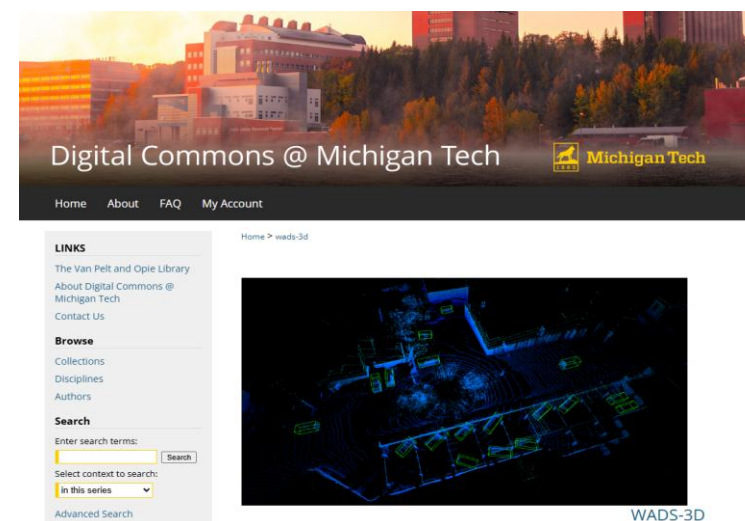
> 45 TB raw data

> 7GB semantic annotation

> 80K downloads

WADS-3D Dataset:

<https://digitalcommons.mtu.edu/wads-3d/>



30,104 labeled cars

High-clutter snowy environments